

Source Documentation

Global Game Jam 2026: Hardware & Software

Contents

1. Introduction	4
1.1. Important Dates and Timeline	4
2. Technology Used	4
2.1. Hardware	4
2.1.1. Raspberry Pi Pico 2 WH	4
2.1.2. Standard Push Buttons	4
2.1.3. 22 Gauge Wire	5
2.1.4. 2.2'' LCD Screen	5
2.1.5. Solder Boards	5
2.1.6. Speakers	5
2.1.7. Audio Interface and Amplifier	5
2.1.8. Total Hardware cost	5
2.2. Software	6
2.2.1. Gb Studio	6
2.2.2. Solidworks/FreeCAD	6
2.2.3. Cura Slicer	6
2.2.4. Git	7
2.2.5. Total Software Cost	7
3. Schematics and Models	7
3.1. Printed Shell	7
3.2. Hardware Internals	7
3.3. Git Workflow	7
4. Prior to Day One	7
4.1. Buying the Hardware	7
4.2. Checking Competition Legality	8
4.3. Preparations	8
5. The Game Jam	8
5.1. Theme	8

5.2. Our Game	8
5.2.1. Art Style	9
5.2.2. Music Style	9
5.2.3. Screenshots	9
5.3. Total Timeline	9
5.3.1. Day 1	9
5.3.2. Day 2	9
5.3.3. Day 3	9
5.3.4. Presentation Time	9
6. Results and Reflection	10
6.1. Software Failures	10
6.2. Hardware Failures	10
6.3. Software Successes	10
6.4. Hardware Successes	10
6.5. Event Failures	10
6.6. Overall Takeaway	10
7. Next Steps	10
7.1. Gameboy Version 2	10
7.2. More Games	10
8. Credits	11

1. Introduction

This documentation serves as the collection of information between the hardware and software of this project.

1.1. Important Dates and Timeline

The entire game was made on a single weekend, from January 30th to February 1st. The hardware construction began 2 weeks earlier. To stay within the nature of the competition, the game can be played, and is presented in a more typical “.exe” format and in a web browser on The Game Jam Website. The Game was finished on Feb. 1st, 2026. We continued to work and develop hardware into the following week.

2. Technology Used

2.1. Hardware

2.1.1. Raspberry Pi Pico 2 WH

We went with the Raspberry Pi Pico 2W as the “brain” of our demonstration. Inspired by the Pico GB project, we wanted to take the core reverse engineering provided by the project and put our own game and modifications onto the project.

2.1.2. Standard Push Buttons

As a cost saving measure, we got a large pack of standard push buttons as all inputs for the hardware.

2.1.3. 22 Gauge Wire

We used this wire because it was available and thicker than the original gauge wire used by Pico GB, any gauge thicker than 24AWG will do.

2.1.4. 2.2" LCD Screen

While not the original model used by Pico GB, we matched their specs exactly, 2.2inch diagonally with 220p x 176p resolution. Any screen with these specs will do. It is important that they have an SD card slot for game reading.

2.1.5. Solder Boards

To save on weight and thickness we used some donated solderboard with a matching connector layout as a breadboard. This allowed us to match our prototyping, cut cost, and save time when building the final model. Any breadboard, or (preferably) solderboard, will do.

2.1.6. Speakers

We went with an 8ohm resistive speaker for sound. Any wattage from 1-2 Watts will do. It should be noted that higher the wattage the louder the sound.

2.1.7. Audio Interface and Amplifier

We went with a standard I2C sound board for all audio interfacing.

2.1.8. Total Hardware cost

Item Cost	
Raspberry Pi Pico 2 WH (2 pack)	\$11.98
Standard Push Buttons (Large bundle)	\$9.99
22 Gauge Wire	\$5.99
2.2" LCD Screen (2 pack)	\$26.50
Solder Boards (3 pack)	\$4.99
Speaker (2 pack)	\$11.98
Audio Interface (3 pack)	\$11.99
Total Cost	\$83.42

Table 1: Total Cost Breakdown

2.2. Software

2.2.1. Gb Studio

This was our go-to game engine for designing, testing, and compiling/presenting our game to the game jam community.

Link: <https://www.gbstudio.dev>

2.2.2. Solidworks/FreeCAD

For all 3D models for the Gameboy cases, we used a variety of solidworks/FREECAD to design, view, and slice in order for us to 3D print what we needed. All models are included in section 3.

Link: <https://www.solidworks.com>

Link: <https://www.freecad.org>

2.2.3. Cura Slicer

For slicing models created for the gameboy cases, we decided to go with cura slicer in order to print. For printers the following were available:

- Creality Ender V3 SE
- Creality Ender V2 Standard

2.2.4. Git

As seen with this repository, git was the version control system the team agreed on. We then uploaded the repository to github.

Link: <https://git-scm.com>

Link: <https://github.com/JEvan234/Global-Game-Jam-2026>

2.2.5. Total Software Cost

All of the software used for this project was free and open source software (FOSS). Outside of solidworks, which has alternatives like FREECAD, none of the software has a purchase or licensing fee.

3. Schematics and Models

3.1. Printed Shell

3.2. Hardware Internals

3.3. Git Workflow

4. Prior to Day One

4.1. Buying the Hardware

We bought our hardware from a variety of storefronts ranging from our local Micro Center to Amazon online.

4.2. Checking Competition Legality

According to the Game Jam Website, link at bottom, All alternative-hardware titles were competition legal and welcome.

Link: <https://globalgamejam.org/jam-sites/2026/kennesaw-state-university>

4.3. Preparations

Leading into day one of the Game Jam:

- Initialized the repository and documentation
- Acquired materials
- Downloaded software
- Assembled team and established goals for the jam

5. The Game Jam

5.1. Theme

The Game jam Theme this year is **Mask**

5.2. Our Game

Veil of Dusk and Dawn is a asymmetrical Co-Op multiplayer title meant to be ran on any hardware capable of gameboy emulation. You play as 1 man split across time as you attempt to return a magical mask back to it's rightful place, reuniting the timeline. One player is in the "Dusk" form, meaning he cannot see with the mask on, but can interact with the world, while in this form the player should be careful to not set off the alarms at every exhibit. To help in navigate, a second player named "Dawn" can see the world, but cannot interact with the world as in his timeline the

mask is already returned. Dawn can use his vision to guide the Dusk player to returning the mask.

5.2.1. Art Style

For this title, we used tilemaps made out of tiles on an 8x8 grid. For exact art style looks, refer to the figure below:

5.2.2. Music Style

5.2.3. Screenshots

Attached are a few game screenshots, order is random and not a reflection of gameplay:

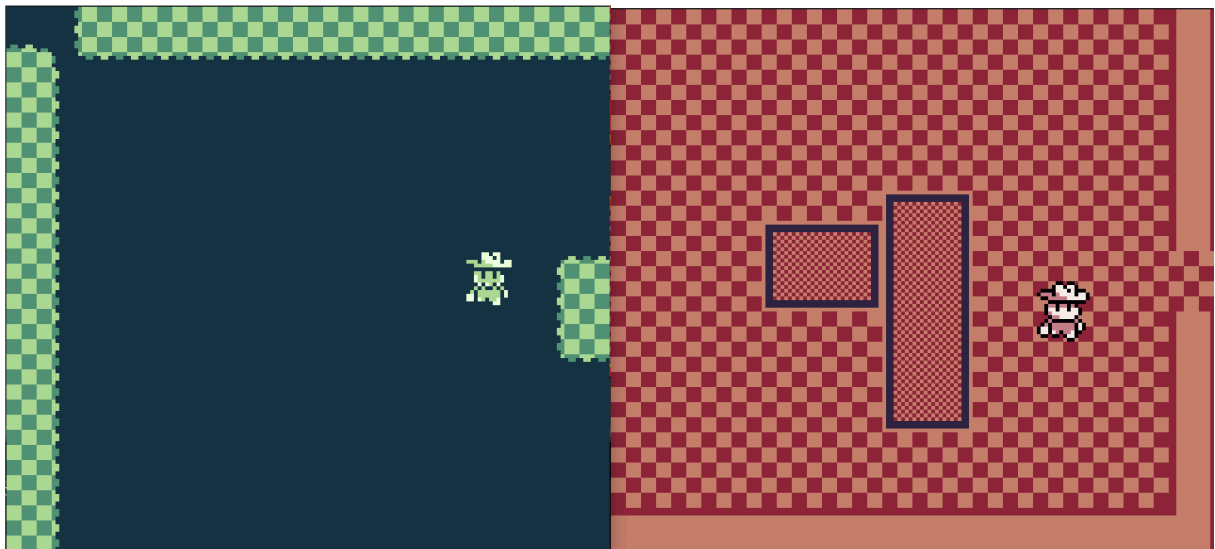


Figure 1: Game Screenshot, Night left, Day right

5.3. Total Timeline

5.3.1. Day 1

5.3.2. Day 2

5.3.3. Day 3

5.3.4. Presentation Time

6. Results and Reflection

6.1. Software Failures

6.2. Hardware Failures

6.3. Software Successes

6.4. Hardware Successes

6.5. Event Failures

6.6. Overall Takeaway

7. Next Steps

Included below are the next steps we are taking to polish the work we completed for the game jam. There are no deadlines, nor promise of completion. All that is included here is a goal for us to continue developing this project into what we initially pictured before the jam.

7.1. Gameboy Version 2

We plan to continue working on the gameboy and primarily redo the outer 3D printed shell to be thinner (using a solderboard instead of breadboard) and utilize a cleaner design.

7.2. More Games

If we have the time a few people might develop more games for our handheld on GBstudio. It is also possible to find other ROMs for it and install them via the SD card.

8. Credits

A special thanks to:

- Mert Acar (Software, Game Logic)
- Oluwajomiloju Adediran (Software, Game Logic)
- Jacob Evans (Hardware, Software, Documentation)
- Raam Patel (Hardware, Documentation)
- Tri Pham (Software, Art, Music)

List of Figures

Table 1: Total Cost Breakdown	6
Figure 1: Game Screenshot, Night left, Day right	9